

ODE-Based Modeling and Comparative Analysis of Lithium-Ion Battery Charging Strategies

MAT292 Course Project

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Abstract

Lithium-ion batteries underpin modern energy storage systems, from electric vehicles to grid-scale applications, making fast and reliable charging strategies increasingly critical. However, aggressive charging can significantly accelerate thermal stress and long-term degradation, while conservative approaches limit practicality. In this project, we develop a reduced-order mathematical model of lithium-ion battery charging based on coupled ordinary differential equations that capture thermal dynamics, transient electrochemical polarization, and degradation driven by solid–electrolyte interphase (SEI) growth. Using this framework, we systematically compare four common charging strategies: constant current (CC), constant-current–constant-voltage (CCCV), CCCV with pulsed current, and constant voltage (CV). Simulation results demonstrate that voltage-controlled strategies achieve the fastest charging times but induce the highest peak temperatures and SEI growth, frequently approaching or exceeding conservative safety limits. In contrast, current-controlled and hybrid CCCV-based strategies offer a more balanced trade-off between charging speed, thermal safety, and degradation. A composite efficiency metric confirms that moderate-to-high current CC and nominal-voltage CCCV strategies provide the most favorable overall performance. These results quantitatively explain the widespread adoption of CCCV charging in commercial battery management systems and illustrate the value of reduced-order ODE models for evaluating and designing charging protocols.

1 Introduction

Lithium-ion batteries are a cornerstone of modern energy infrastructure, powering consumer electronics, electric vehicles (EVs), and large-scale renewable energy storage. Global demand is surging, driven by rapid decarbonization efforts. In 2023 alone, approximately 14 million new EVs were sold worldwide—a 40% increase in battery demand—while global grid storage capacity surpassed 85 GW [1, 2]. Concurrently, a nearly 90% cost reduction since 2010 has accelerated widespread adoption [2].

Despite this progress, battery charging remains a critical bottleneck. Aggressive fast-charging protocols reduce charging time but accelerate degradation, leading to capacity loss and reduced cycle life. Conversely, overly conservative strategies preserve longevity at the expense of practicality, especially for time-sensitive EV charging. Bridging this trade-off is essential: studies indicate that optimized charging can extend battery lifetime by up to 30% while maintaining, or even improving, charging speed [3, 4].

Motivated by this need, this project develops a computational model of lithium-ion battery charging to enable the systematic comparison of different strategies. Rather than relying on empirical testing, we construct a simplified model using differential equations to represent key physical processes, including state of charge, thermal dynamics, and degradation. This approach balances physical accuracy with computational efficiency, allowing us to simulate how charging profiles influence internal stress, heat generation, and long-term capacity fade. The ultimate goal is to provide a clear, mechanism-based analysis of common charging protocols, offering insight to inform strategies that better balance speed, efficiency, and battery health.

2 Mathematical Foundation

To capture the dominant physical processes governing lithium-ion battery charging, we model the system as a set of coupled ordinary differential equations. Each mechanism is implemented

as an independent module contributing to the evolution of a shared battery state vector,

$$\mathbf{y}(t) = [V, I, R, T, S, D, V_c], \quad (1)$$

where V is the terminal voltage, I is the current, R is the internal resistance, T is the battery temperature, S is the state of charge, D represents cumulative degradation due to solid-electrolyte interphase (SEI) growth, and V_c is the transient polarization voltage. This modular structure allows individual physical effects to be isolated, analyzed, and modified while remaining fully coupled through shared state variables.

2.1 Thermal Effects

Thermal behavior plays a critical role in battery performance, safety, and lifetime, as elevated temperatures accelerate electrochemical reaction rates and parasitic side reactions, increasing the risk of degradation and thermal runaway during charging [5]. To capture these effects, battery temperature dynamics are modeled using a lumped-parameter approach that assumes a spatially uniform cell temperature. Heat is generated internally due to resistive (Joule) heating when current flows through the battery, while simultaneous heat exchange with the environment acts to cool the cell toward ambient conditions. Combining internal heat generation and environmental cooling yields the temperature evolution

$$\frac{dT}{dt} = \frac{I^2 R}{mc} - k(T - T_{\text{amb}}), \quad (2)$$

where I is the battery current, R is the internal resistance, m is the battery mass, c is the specific heat capacity, k is an effective cooling constant, and T_{amb} is the ambient temperature. This formulation captures the dominant thermal effects during charging and highlights the strong coupling between charging protocols, temperature rise, and long-term battery health [5, 7, 8].

2.2 Transient Dynamics

During charging, lithium-ion batteries exhibit transient voltage deviations from equilibrium due to electrochemical polarization. These effects

arise from finite charge-transfer kinetics at the electrode–electrolyte interface, ion diffusion limitations within electrode materials, and the formation of electrical double layers [6, 8].

To capture this behavior, polarization is modeled using a first-order RC circuit representation, a widely adopted reduced-order approach in battery charging analysis [8, 9]. The transient polarization voltage $V_c(t)$ evolves according to

$$\frac{dV_c}{dt} = -\frac{1}{RC}V_c + \frac{1}{C}I(t), \quad (3)$$

where R represents the polarization resistance and C is the effective double-layer capacitance.

The first term models exponential relaxation of polarization when current decreases or stops, while the second term represents voltage buildup under sustained current flow. The system response time is governed by the RC time constant $\tau = RC$. Physically, this formulation captures lithium-ion accumulation near electrode surfaces during aggressive charging and subsequent relaxation during rest periods. This mechanism is particularly relevant for pulse charging, negative pulse charging, and sinusoidal charging strategies, which intentionally introduce current variation to mitigate polarization and improve charging efficiency [9, 10, 11].

2.3 Long-Term Degradation (SEI Growth)

Long-term degradation in lithium-ion batteries is dominated by the growth of the solid–electrolyte interphase (SEI) layer on the anode. Although the SEI initially stabilizes the electrode–electrolyte interface, continued growth irreversibly consumes cyclable lithium and increases internal resistance, leading to capacity fade and reduced cycle life [7].

In this model, degradation is represented by a cumulative SEI growth variable $D(t)$, whose evolution is governed by a first-order ordinary differential equation. Consistent with simplified SEI models in the literature, the SEI growth rate is assumed to depend on battery temperature and charging aggressiveness rather than detailed microscopic reaction pathways [7]. The governing

equation takes the form

$$\frac{dD}{dt} = k_{\text{SEI}} f(I, V_c) \exp\left(-\frac{E_a}{R_g T}\right), \quad (4)$$

where k_{SEI} is a scaling constant for SEI growth, $f(I, V_c)$ is a monotonically increasing function of charging current I and transient polarization voltage V_c , E_a is an effective activation energy, R_g is the universal gas constant, and T is the battery temperature.

This formulation reflects two well-established physical drivers of SEI formation. First, SEI growth is thermally activated, with higher temperatures accelerating parasitic side reactions at the anode surface. This Arrhenius-type dependence is widely reported in experimental and modeling studies and captures the strong sensitivity of degradation rates to temperature [5, 7]. Second, aggressive charging conditions increase electrochemical overpotential and lithium-ion concentration gradients, which promote non-ideal electrolyte reduction reactions that contribute to SEI thickening [7].

In the numerical implementation, this equation is evaluated incrementally and integrated over time, allowing irreversible degradation to accumulate across charging cycles. Because temperature and polarization dynamics are governed by separate state equations, SEI growth is indirectly coupled to thermal effects and transient electrochemical behavior. As a result, charging strategies that reduce internal heating or allow polarization relaxation—such as pulse charging, negative pulse charging, or sinusoidal current profiles—naturally lead to slower degradation rates in the model [8, 9, 10, 11].

3 Methodology

We developed a mathematical model that integrates the previously described mechanism equations to simulate the evolution of battery state variables during charging. The system state is updated through a combination of algebraic relations and numerical integration of coupled ordinary differential equations. Among the tracked variables, charging speed and cumulative SEI

growth are used as primary metrics to evaluate the performance of each charging policy.

3.1 Numerical Integration

The coupled differential equations governing thermal dynamics, state of charge, transient polarization, and degradation are solved numerically using the classical fourth-order Runge–Kutta (RK4) method. RK4 was selected due to its favorable balance between accuracy and computational efficiency. Its fourth-order convergence provides high numerical precision, while its stability properties make it suitable for moderately stiff systems commonly encountered in battery dynamics.

At each time step, RK4 evaluates the system derivatives at four intermediate points and combines them through weighted averaging to advance the solution. Compared to first-order methods such as forward Euler integration, RK4 significantly reduces both local and global truncation errors while remaining computationally tractable for multi-cycle charging simulations.

3.2 Charging Policy Definitions

We evaluate 4 distinct charging methods, each defined by combinations of Constant Current (CC), Constant Voltage (CV), and pulse-based charging. Individual charging policies are parameterized by a prescribed current level, voltage setpoint, and, where applicable, a pulse frequency and duty cycle. For composite strategies, such as CC–CV charging, the applied control law transitions between modes once a specified state-of-charge (SOC) threshold is reached.

3.3 Parameter Selection and Fair Comparison

Because charging strategies respond differently to parameter choices, direct comparison using identical control parameters would bias the results. To ensure fairness across charging methods, we test three reasonable parameter sets for each strategy and select the configuration that yields an equivalent charging time to reach 100% SOC. This normalization ensures that observed differences in thermal behavior and degradation arise from the charging profile itself rather than from disparities in total charging duration.

To maintain physical realism, an upper temperature constraint of 60°C is imposed during parameter selection. This limit reflects commonly recommended operating thresholds for commercial lithium-ion batteries and prevents the inclusion of unsafe charging conditions [12].

3.4 Cycle Simulation and Degradation Tracking

Each charging policy is simulated over ten consecutive charging cycles. At the start of each cycle, the battery state variables are reset to their initial values, while cumulative SEI degradation is retained. Although real batteries experience hundreds or thousands of cycles, we observe that SEI growth per cycle remains approximately constant under fixed operating conditions. As a result, ten cycles are sufficient to resolve relative differences in degradation rates between charging strategies.

3.5 Composite Performance Metric

To compare charging strategies across multiple objectives, including charging time, peak temperature, and cumulative SEI growth, we define a composite efficiency metric. Each performance measure is normalized across all charging methods to account for differences in scale and units. The normalized metrics are then inverted where appropriate (so that lower temperature and lower degradation correspond to higher scores) and combined using equal weighting to produce a single efficiency score.

This composite metric provides a consistent basis for ranking charging strategies by their overall balance between speed, thermal safety, and degradation, rather than by any single performance criterion.

4 Results

Four lithium-ion charging policies—Constant Current (CC), Constant-Current–Constant-Voltage (CCCV), CCCV with pulsed current (CCCV-Pulse), and Constant Voltage (CV)—were evaluated under a unified simulation framework. All simulations used identical battery parameters and termination conditions to ensure a fair comparison. Constant-current charging was tested at 3 A, 6 A, and 9 A (approximately 1C–3C for

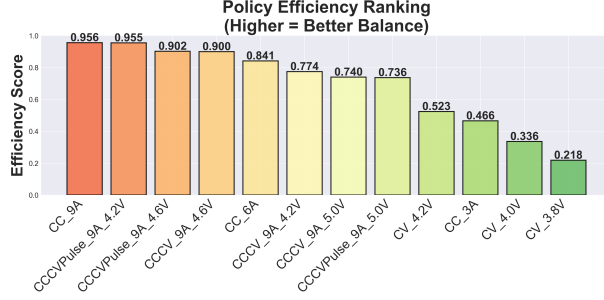


Figure 1: Overall performance score (E-score) for each charging strategy, combining charging time, peak temperature, and cumulative SEI growth. Lower values indicate superior overall performance.

a 3 Ah cell), while voltage-controlled strategies were evaluated at 4.2 V, 4.6 V, and 5.0 V to span nominal to aggressive operating regimes.

Performance was assessed using three primary metrics: cumulative solid-electrolyte interphase (SEI) growth over ten charging cycles, peak battery temperature during charging, and time required to reach 100% state of charge (SOC). These metrics capture the key trade-offs between degradation, thermal safety, and charging speed.

To provide an overall comparison, these metrics were combined into a normalized performance score (E-score), where lower values indicate better overall charging performance.

Figure 1 highlights the dominant trade-offs between the evaluated charging methods. Pure constant-voltage charging consistently performs worst due to excessive thermal and degradation penalties, despite its fast charging times. Constant-current charging achieves stable but slower performance, while CCCV-based strategies occupy a favorable intermediate regime with balanced efficiency, thermal safety, and degradation control.

Cumulative SEI growth across ten charging cycles is shown in Fig. 2. Increasing either charging current or applied voltage leads to higher SEI accumulation, indicating accelerated degradation under more aggressive charging conditions. Voltage-controlled strategies, particularly pure CV charging at elevated voltages, exhibit the greatest SEI growth.

Peak battery temperatures reached during

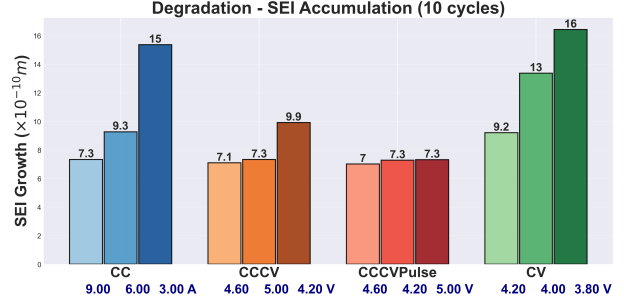


Figure 2: Cumulative solid-electrolyte interphase (SEI) growth over ten charging cycles for each charging policy.

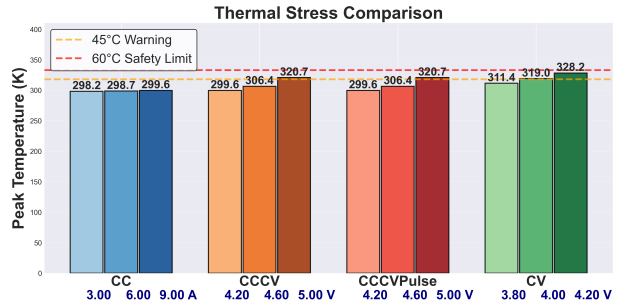


Figure 3: Peak battery temperature reached during charging for each policy. Reference lines indicate typical safety and performance thresholds.

charging are presented in Fig. 3. For all strategies, higher current or voltage results in increased thermal stress. Pure CV charging produces the highest peak temperatures and frequently exceeds conservative safety thresholds, especially at applied voltages above 4.2 V. In contrast, CC charging exhibits a more gradual temperature increase and remains within acceptable limits even at higher C-rates. CCCV and CCCV-Pulse strategies reduce peak temperatures relative to pure CV charging by limiting early-stage current.

Charging times required to reach full SOC are summarized in Fig. 4. Increasing the charging voltage yields the largest reduction in charging time, particularly above the nominal 4.2 V terminal voltage. Pure CV charging achieves the shortest charging times, whereas CC charging exhibits a near-linear reduction in time with increasing current. CCCV-based strategies maintain competitive charging times while avoiding extreme operating conditions.

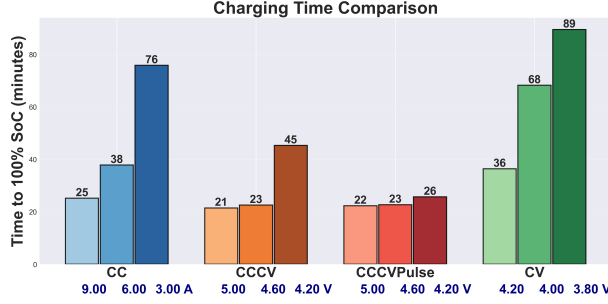


Figure 4: Time required to reach 100% state of charge for each charging policy and parameter set.

5 Discussion

The observed trends can be explained by the underlying physical mechanisms governing lithium-ion battery behavior during charging. A central insight is that both peak temperature and SEI growth are driven primarily by ohmic heating, which scales with the square of the charging current. As a result, effective charging strategies must regulate not only the magnitude of the current but also its temporal profile.

Figure 5 shows the open-circuit voltage (OCV) curve used in the model, illustrating the intrinsic SOC–voltage relationship of the battery.

Pure constant-voltage charging performs poorly from a thermal perspective because it imposes a large voltage mismatch at low SOC. This mismatch drives excessive current early in the charge, resulting in sharp temperature increases and accelerated SEI growth. Although CV charging minimizes total charging time, the associated thermal stress often exceeds conservative safety limits. In contrast, constant-current charging maintains a controlled and predictable current profile. Even at elevated C-rates, CC charging limits peak temperatures and exhibits a near-linear trade-off between charging speed and thermal stress. This behavior reflects direct control over the primary driver of heat generation.

Hybrid CCCV strategies combine the advantages of both approaches. The initial current-limited phase enables rapid energy transfer while preventing excessive early heating, and the subsequent voltage-limited phase allows the charge to

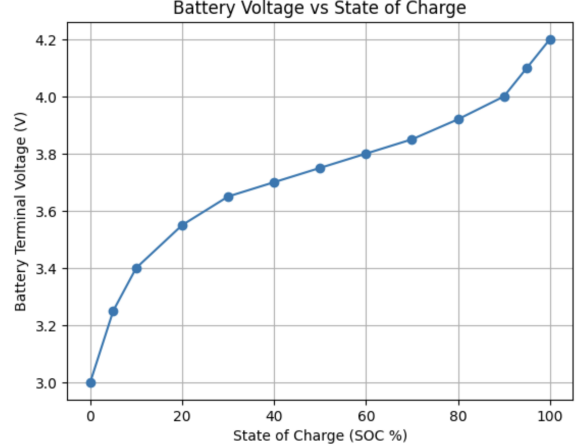


Figure 5: Battery open-circuit voltage as a function of state of charge (SOC) used in the model.

complete safely. Pulse-assisted variants provide modest reductions in peak temperature but do not significantly reduce cumulative SEI growth.

Overall, charging performance is more sensitive to voltage selection than to current magnitude. Operating close to the battery’s intrinsic SOC–voltage curve minimizes unnecessary voltage mismatch and reduces mean-square current, thereby limiting both thermal stress and degradation. CCCV charging naturally satisfies this criterion, explaining its strong balance between charging speed, safety, and longevity.

6 Assumptions and Limitations of the Model

While the proposed model captures the dominant electrical, thermal, and degradation mechanisms relevant to fast charging, it relies on simplifying assumptions to remain computationally tractable. These assumptions are common in reduced-order battery models but limit the interpretation of absolute results and long-term predictions.

6.1 Lumped Thermal Representation

The battery is modeled as a single lumped element with spatially uniform voltage, temperature, state of charge, and degradation state. This approach neglects internal gradients in temperature, lithium concentration, and electric potential that

arise in real cells, particularly under high-current or pulsed charging conditions [14]. As a result, localized hot spots and regions of accelerated degradation are not resolved, and peak thermal stress and SEI growth may be underestimated for aggressive charging strategies [5, 8].

6.2 Separation of Algebraic and Differential Dynamics

In the numerical implementation, electrical variables such as current, resistance, and open-circuit voltage are updated algebraically, while thermal and degradation states evolve dynamically. This separation assumes instantaneous electrical equilibrium relative to slower processes, simplifying computation and improving numerical stability. However, it neglects inductive effects, finite charge-transfer kinetics, and tight coupling between fast electrical transients and slower dynamics. Consequently, sharp current spikes and voltage transients are smoothed out, potentially underrepresenting short-timescale stresses associated with highly dynamic charging profiles.

6.3 Semi-Empirical SEI Growth Model

Long-term degradation is represented using a semi-empirical SEI growth model driven by temperature, state of charge, and applied current. While this formulation captures the qualitative dependence of degradation on thermal stress and charging aggressiveness, it does not resolve detailed electrochemical reaction pathways or distinguish between reaction-limited and diffusion-limited growth regimes. In addition, SEI growth does not feed back into capacity fade or resistance increase, limiting its influence on subsequent charging behavior. As a result, the predicted SEI evolution should be interpreted primarily as a relative indicator of degradation severity rather than an absolute measure of long-term aging.

Taken together, these limitations indicate that the model is best suited for comparative analysis of charging strategies. While extreme internal conditions and long-term aging effects may be underestimated, all strategies are evaluated under

consistent assumptions, ensuring that relative performance trends and trade-offs remain robust within the scope of the simulation.

7 Future Work

Several extensions would significantly improve the accuracy and applicability of the present model. First, extending simulations to longer cycle counts would help determine whether early-cycle advantages observed for optimal charging strategies persist over hundreds or thousands of cycles, as required in practical applications such as electric vehicles [1, 2].

Second, the model is well suited for parameter calibration to represent different battery chemistries, geometries, and operating environments. Because the governing equations are intentionally abstracted, key parameters such as SEI growth rates and thermal coefficients can be tuned to match application-specific requirements, enabling adaptation to both EV fast charging and stationary grid-scale energy storage systems [7, 8]. Higher-fidelity submodels may also be incorporated for selected mechanisms while preserving the overall framework.

Third, future work could incorporate multi-objective optimization, allowing charging time, thermal stress, and degradation to be weighted differently depending on application priorities. For example, EV charging emphasizes speed within safety constraints, whereas grid storage prioritizes longevity and efficiency [8].

Finally, adaptive and feedback-based charging strategies represent a promising extension. Rather than relying on fixed charging profiles, future implementations could dynamically adjust current or voltage in response to real-time measurements of temperature, state of charge, or degradation indicators. Such approaches align closely with modern battery management systems and would improve both predictive accuracy and practical relevance [3].

Ultimately, model fidelity should be enhanced by strengthening the coupling between electrical, thermal, and aging dynamics—particularly by linking SEI growth directly to resistance in-

crease and capacity fade—and by validating predictions against experimental fast- and pulse-charging data. These improvements would evolve the framework from a trend-analysis tool into a modular, industry-relevant platform for charging strategy design.

8 Conclusion

In this project, we developed a reduced-order mathematical model to study lithium-ion battery charging behavior under a range of commonly used charging strategies. By formulating the charging process as a system of coupled ordinary differential equations, the model captures the interplay between electrical dynamics, thermal effects, transient polarization, and long-term degradation driven by solid–electrolyte interphase (SEI) growth. This framework enables a systematic and mechanism-based comparison of charging protocols under consistent operating conditions.

Through numerical simulation, we evaluated constant-current, constant-voltage, and hybrid charging strategies with respect to charging speed, thermal stress, and cumulative degradation. The results highlight a fundamental trade-off between fast charging and battery health: more aggressive voltage-controlled strategies significantly reduce charging time but induce elevated temperatures and accelerated degradation, while current-controlled strategies offer improved thermal safety at the cost of longer charging durations. Hybrid CCCV-based strategies emerge as the most balanced approach, combining rapid early charging with effective thermal moderation, which explains their widespread adoption in commercial battery management systems.

Beyond specific performance outcomes, this work demonstrates the utility of ordinary differential equation models for analyzing and comparing battery charging strategies. While the model is intentionally simplified, it provides clear insight into how charging profiles influence internal battery stress and degradation trends. Because all strategies are evaluated under the same assumptions, the conclusions drawn are robust for relative comparison, even if absolute performance

predictions are limited.

Overall, this study illustrates how reduced-order ODE-based models can serve as effective tools for guiding charging strategy design and evaluation. By balancing physical interpretability with computational efficiency, such models offer a valuable foundation for future optimization and control-oriented extensions aimed at improving charging speed while preserving battery safety and longevity.

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A Governing Equations of the Battery Model

A.1 State Vector

The battery state is represented by

$$\mathbf{y}(t) = \begin{bmatrix} V \\ I \\ R \\ T \\ \text{SOC} \\ \text{SEI} \\ V_c \end{bmatrix},$$

where

- V : open-circuit voltage (V),
- I : current into the cell (A),
- R : internal resistance (Ω),
- T : cell temperature (K),
- SOC : state of charge (0–1),
- SEI : SEI layer thickness (arb. units),
- V_c : transient polarization voltage (V).

The full system is written as a sum of individual physical mechanisms:

$$\frac{d\mathbf{y}}{dt} = \sum_m \left(\frac{d\mathbf{y}}{dt} \right)_m.$$

A.2 Charging Dynamics (State of Charge)

The state of charge evolves according to

$$\frac{d\text{SOC}}{dt} = \frac{I}{C_{\text{nom}} 3600},$$

where C_{nom} is the nominal battery capacity in ampere-hours.

A.3 Thermal Model

Battery temperature changes due to ohmic heating and cooling to the environment:

$$\frac{dT}{dt} = \frac{I^2 R}{mc} - k(T - T_{\text{amb}}),$$

where m is the battery mass, c the specific heat capacity, k the cooling constant, and T_{amb} the ambient temperature.

A.4 Transient RC Polarization Model

Electrochemical polarization is modeled using a first-order RC circuit:

$$\frac{dV_c}{dt} = -\frac{1}{R_c C_c} V_c + \frac{1}{C_c} I,$$

where R_c and C_c denote the polarization resistance and capacitance.

A.5 SEI Growth Models

A.5.1 Full SEI Model

The temperature-dependent rate constant follows an Arrhenius relation:

$$k(T) = k_0 \exp\left(-\frac{E_a}{R_g T}\right),$$

with $R_g = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

The electrochemical stress function is defined as

$$f(\text{SOC}, I, V) = A \exp[\gamma (U_{\text{ref}} - V)] (1 + \beta |I|)^\nu.$$

The resulting SEI growth equation is

$$\frac{d\text{SEI}}{dt} = k(T) f(\text{SOC}, I, V).$$

A.5.2 Simplified SEI Model

A reduced empirical formulation is also considered:

$$\frac{d\text{SEI}}{dt} = k_0 \exp\left(-\frac{E_a}{R_g T}\right) (1 + 2 \text{SOC}^2) (1 + 0.1 |I|).$$

A.6 Algebraic State Updates

The following quantities are updated algebraically at each time step.

Current (Ohm's Law with Polarization)

$$I = \frac{V_{\text{src}} - V - V_c}{R}.$$

Temperature-Dependent Resistance

$$R(T) = R_0 \exp\left[\frac{E_a}{k_b} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right],$$

where k_b is the Boltzmann constant expressed in eV/K.

Open-Circuit Voltage

$$V = \text{OCV}(\text{SOC}),$$

obtained by linear interpolation of empirical SOC-voltage data.

A.7 Charging Policies

Constant Voltage (CV)

$$V_{\text{src}} = V_{\text{set}}.$$

Constant Current (CC)

$$V_{\text{src}} = I_{\text{set}}R + V.$$

CC–CV Hybrid

$$V_{\text{src}} = \begin{cases} I_{\text{set}}R + V, & V < V_{\text{set}}, \\ V_{\text{set}}, & V \geq V_{\text{set}}. \end{cases}$$

Pulsed Charging

$$V_{\text{src}}(t) = \begin{cases} \min(V_{\text{set}}, I_{\text{set}}R + V), & t \bmod (t_{\text{on}} + t_{\text{off}}) < t_{\text{on}}, \\ V, & \text{otherwise.} \end{cases}$$

A.8 Compact System of ODEs

The system integrated numerically is

$$\begin{aligned} \dot{V} &= 0, \\ \dot{I} &= 0, \\ \dot{R} &= 0, \\ \text{SOC} &= \frac{I}{C_{\text{nom}} 3600}, \\ \dot{T} &= \frac{I^2 R}{mc} - k(T - T_{\text{amb}}), \\ \text{SEI} &= (\text{model dependent}), \\ \dot{V}_c &= -\frac{1}{R_c C_c} V_c + \frac{1}{C_c} I. \end{aligned}$$